

$\hat{\mathbf{A}}$

$$x \stackrel{\$}{\leftarrow} Z_q^*$$
$$X = g^x$$

 $\hat{\mathbf{B}}$

$$y \stackrel{\$}{\leftarrow} Z_q^*$$
$$Y = g^y$$

 X Y

$$SK_A = \mathsf{H}(Y^x)$$

$$SK_B = \mathsf{H}(X^y)$$